



جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

Egypt University of Informatics
Computing and Information Sciences
Data Analysis Course

Studying Habits and Academic Performance Analysis

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Introduction

Academic performance is influenced by many factors including study habits, sleep patterns, and classroom attendance. Understanding how these factors interact can help students improve their learning strategies and academic outcomes. In recent years, researchers and educators have shown increasing interest in identifying the habits that contribute most effectively to academic success. This study aims to explore the relationship between students' studying habits and their academic performance using data collected through a survey. By analyzing variables such as weekly study hours, sleep hours, attendance levels, and preferred study methods, the study attempts to identify patterns that may influence students' Grade Point Average (GPA). The results of this analysis provide insights into how different studying behaviors may relate to academic achievement and whether certain habits appear more beneficial than others.

Research Question

How do studying habits such as study hours, sleep hours, and attendance relate to students' academic performance measured by GPA?

Hypothesis

H0 (Null Hypothesis):

There is no relationship between students' study hours per week and their GPA.

H1 (Alternative Hypothesis):

Students who study more hours per week tend to have higher GPA.

Methodology

Population of Interest:

The population of interest for this study is university students who are currently enrolled in undergraduate programs. The survey aims to understand the studying habits and academic performance of students in higher education environments.

Sampling Method:

The sampling method used in this study is convenience sampling. The survey was distributed online using Google Forms and shared with university students through WhatsApp groups that were readily accessible. This method was chosen because it allows quick data collection from a group of students who represent the target population. While convenience sampling may not perfectly represent the entire population of university students, it provides a practical way to collect real-world data within the limited timeframe for this project.

Bias Identification:

Several potential sources of bias were considered during the design of the survey. One possible bias is response bias, where participants may provide answers that they believe are socially desirable rather than completely accurate. To reduce this effect, the survey questions were designed to be neutral and non-leading. Another potential bias is sampling bias, since the survey was distributed primarily among university students who were easily reachable. This may limit the generalizability of the results to the broader student population. Additionally, care was taken to ensure that questions were clearly worded and did not influence participants toward specific answers.

Survey Questions:

How many hours do you study per week on average?

What is your GPA? (Anonymous)

How often do you attend your lectures?

Do you study regularly (daily or almost daily)?

When do you usually start studying before an exam?

What is your preferred study method?

How many hours do you sleep per day?

Do you currently have a job?

Online survey link: https://docs.google.com/forms/d/e/1FAIpQLSfb-mpcgLfWEtbQpupdrWeLzVRVXdemiU1Bswpz_2rIAC7mjA/viewform?pli=1

Number of samples collected: 100

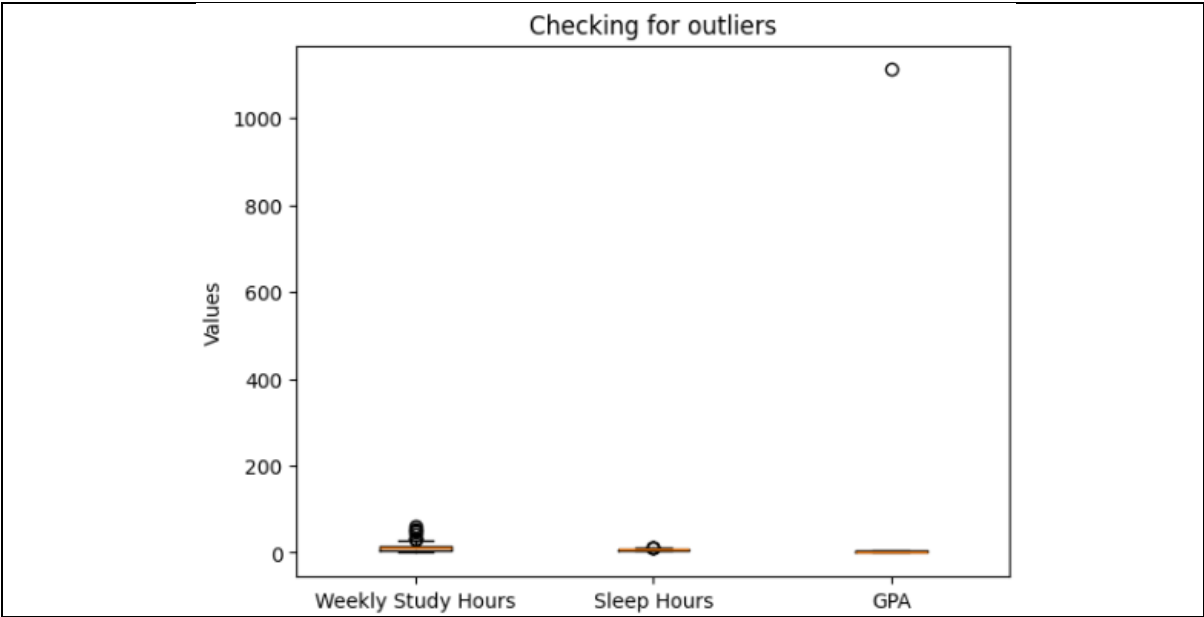
Analysis

To analyze the data collected from this survey, we will calculate basic descriptive statistics such as the mean, median, and mode.

We will also create visual representations of the data using charts and graphs to help identify any trends or patterns.

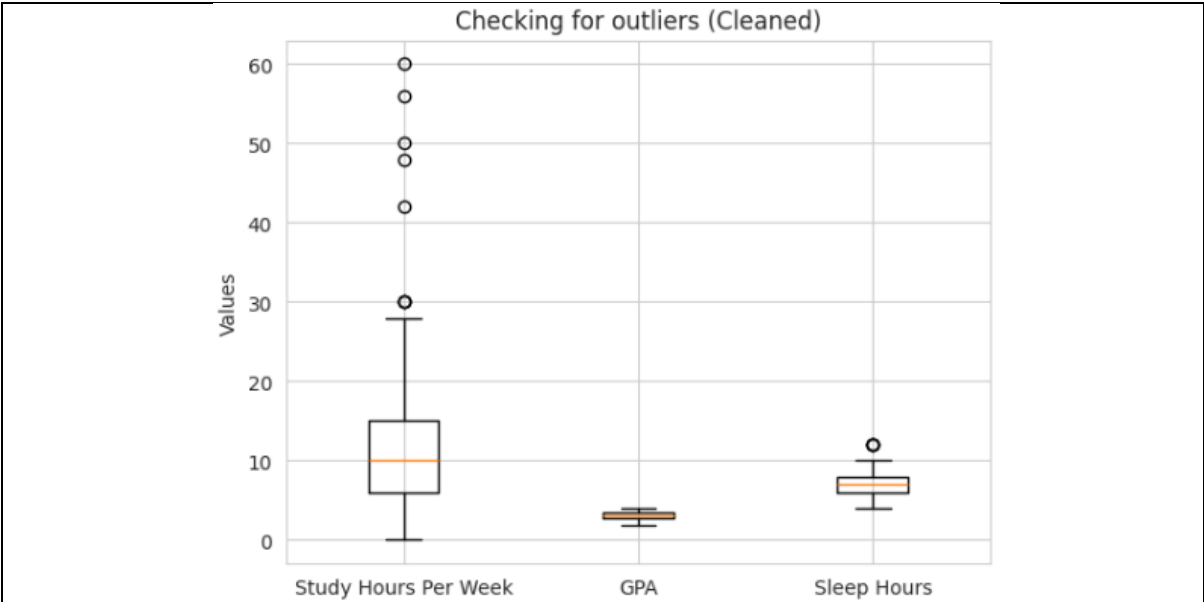
Part I: Quantitative Analysis

Figure 1: Boxplot for Detecting Outliers in Study Hours, Sleep Hours, and GPA



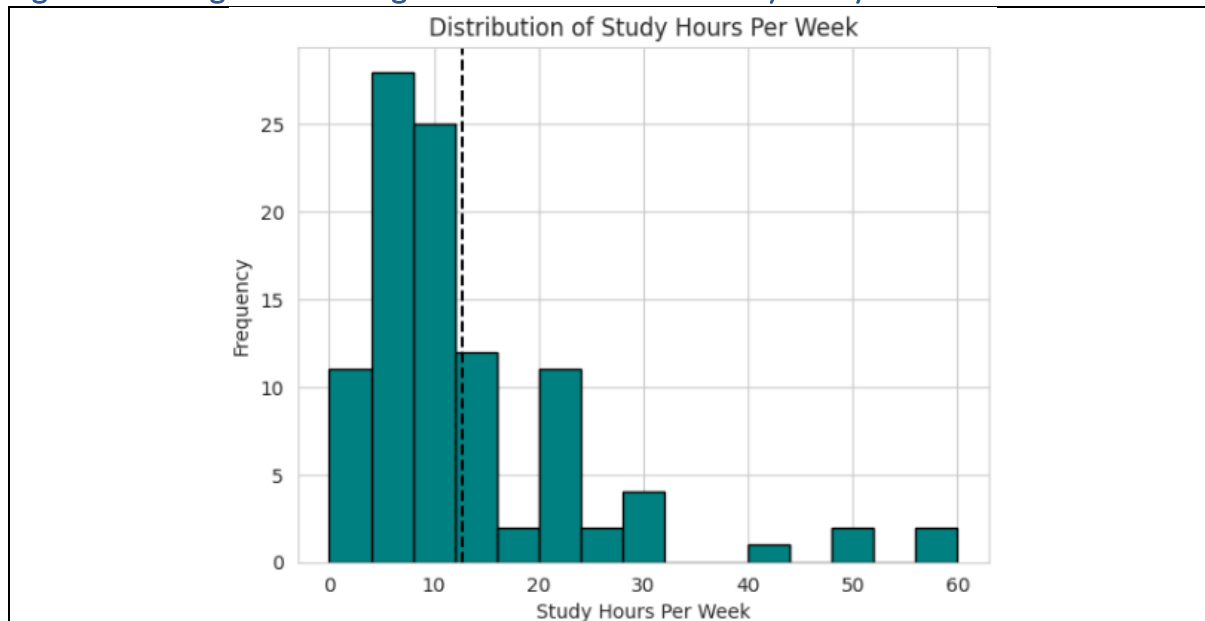
The boxplot was used to identify potential outliers in the quantitative variables, including weekly study hours, sleep hours, and GPA. Most of the values appear within a normal range; however, a very large outlier is visible in the GPA variable. This value is clearly unrealistic because GPA normally ranges between 0 and 4, indicating that the value was likely a data entry error. After identifying this issue, the incorrect value was corrected during the data cleaning process to ensure the dataset remained accurate for further analysis.

Figure 2: Boxplot After Data Cleaning



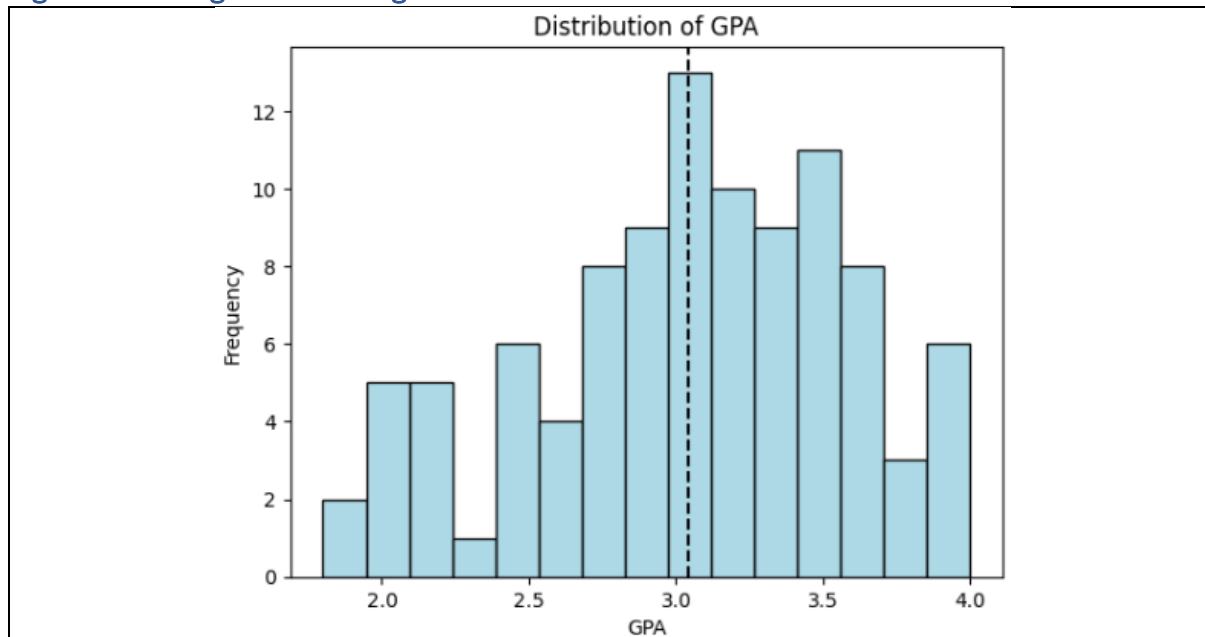
This boxplot shows the distribution of the quantitative variables after the data cleaning process was completed. The unrealistic GPA value identified earlier was corrected, and the dataset now shows values that fall within expected ranges. Some higher values in weekly study hours are still visible, but these are reasonable and represent students who study significantly more than others. Overall, the cleaned data provides a more accurate representation of students' study habits, sleep hours, and academic performance.

Figure 3: Histogram Showing the Distribution of Weekly Study Hours



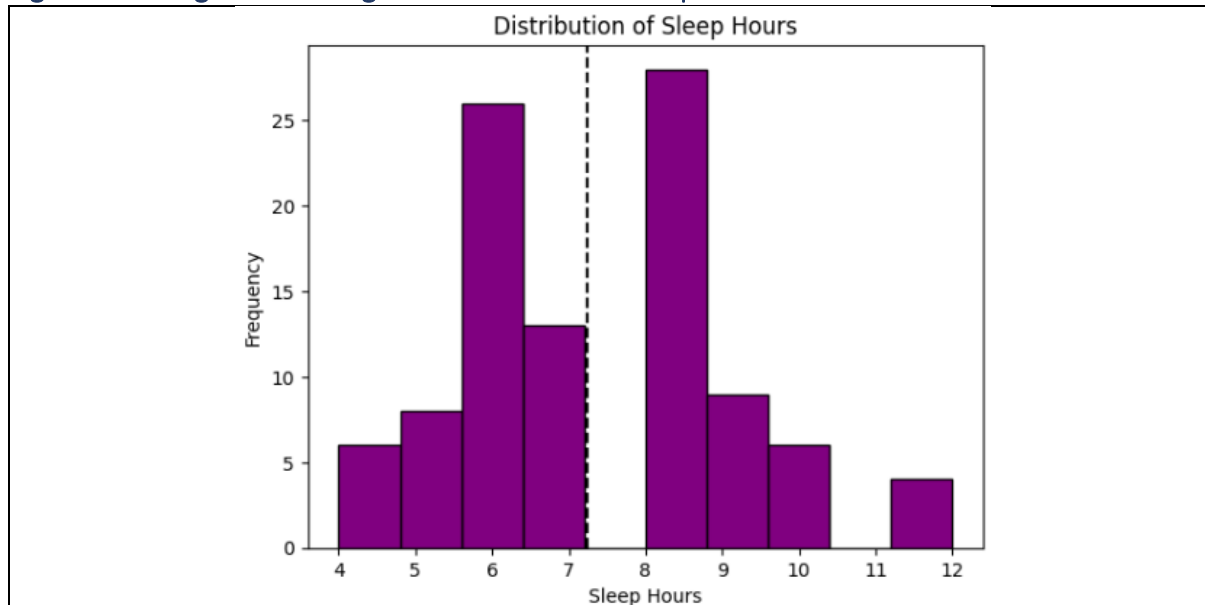
This histogram shows how many hours students study per week based on the survey responses. Most students appear to study between approximately **5 and 15 hours per week**, indicating that this range is the most common study pattern among the participants. A few students report studying significantly more hours, which creates a small right tail in the distribution. The dashed vertical line represents the **average study hours**, which is slightly above 10 hours per week.

Figure 4: Histogram Showing the Distribution of GPA



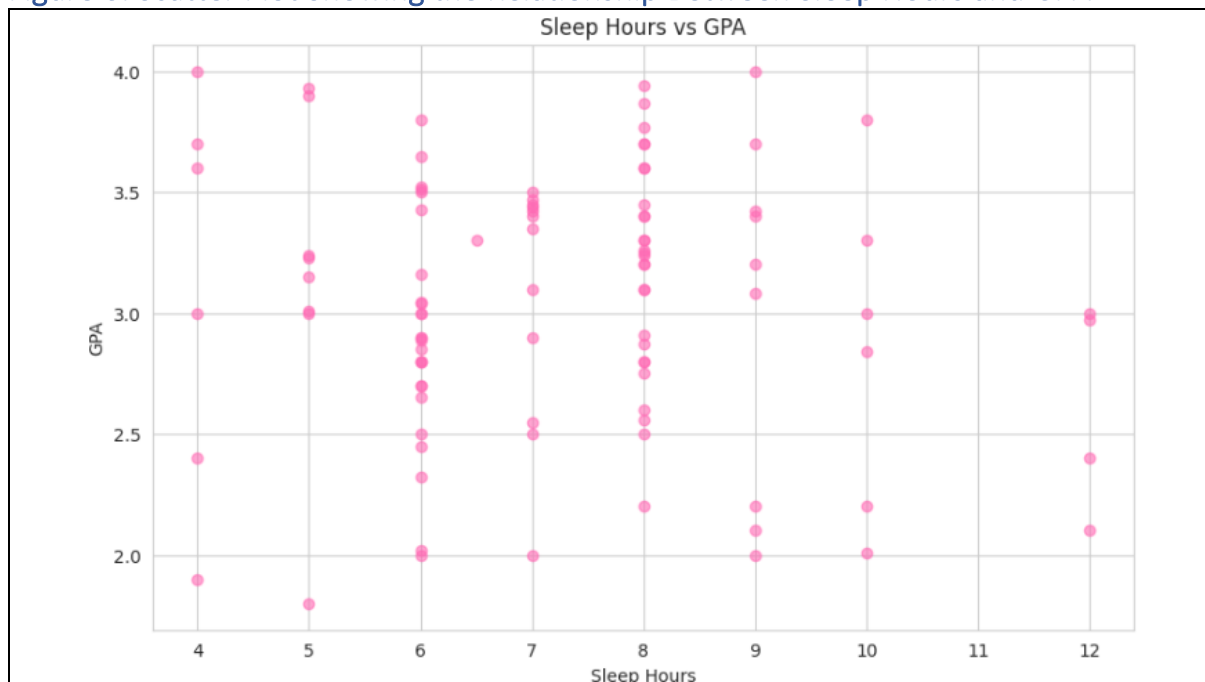
This histogram shows how GPA values are distributed among the surveyed students. Most GPA values appear to fall between approximately **2.5 and 3.5**, indicating that many students in the sample have moderate to relatively strong academic performance. The dashed vertical line represents the **average GPA**, which is around 3.0. The distribution appears fairly balanced, suggesting that the dataset includes students with a range of academic performance levels.

Figure 5: Histogram Showing the Distribution of Sleep Hours



This histogram illustrates the distribution of the number of hours students sleep per night. Most students appear to sleep between **6 and 9 hours**, which is generally considered a typical range for university students. The dashed vertical line represents the **average sleep duration**, which is around 7–8 hours. The distribution suggests that while most students maintain a moderate sleep schedule, a smaller number of students report sleeping significantly more or less than the average.

Figure 6: Scatter Plot Showing the Relationship Between Sleep Hours and GPA



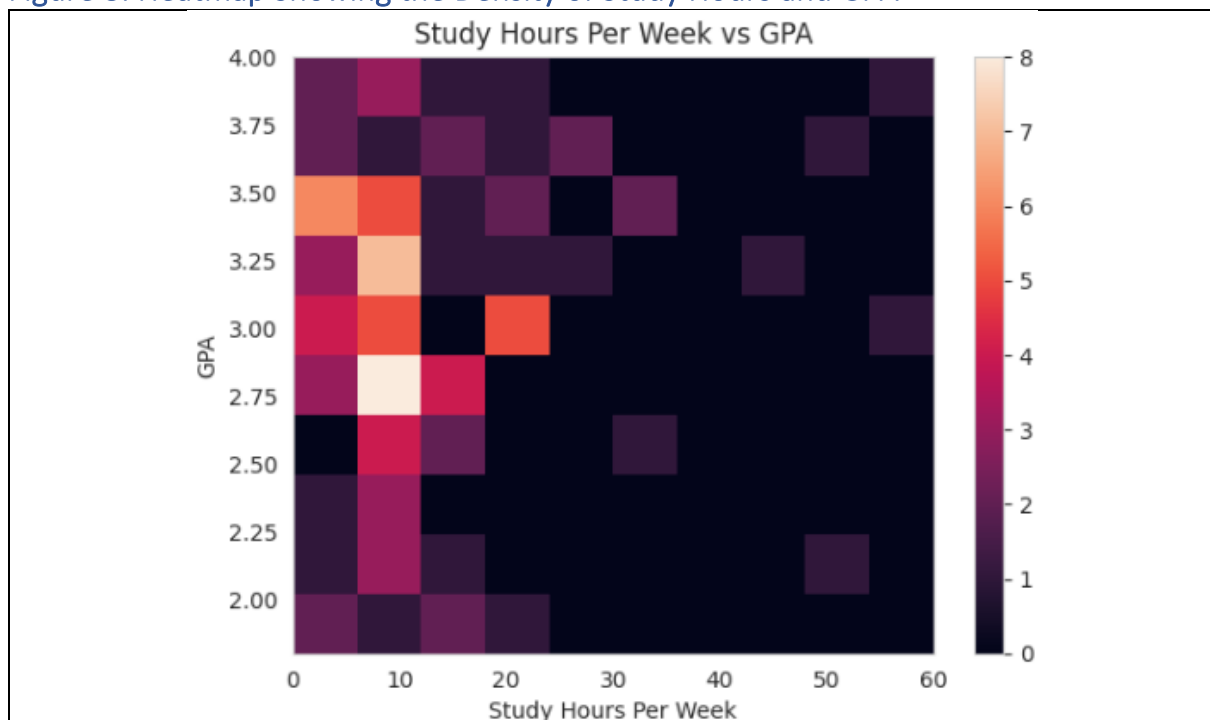
This scatter plot illustrates the relationship between the number of hours students sleep and their GPA. The data points appear widely scattered, indicating that there is **no clear linear relationship** between sleep hours and academic performance in this dataset. Students with both higher and lower GPAs are distributed across different sleep durations. This suggests that sleep hours alone may not strongly predict students' GPA.

Figure 7: Scatter Plot Showing the Relationship Between Weekly Study Hours and GPA



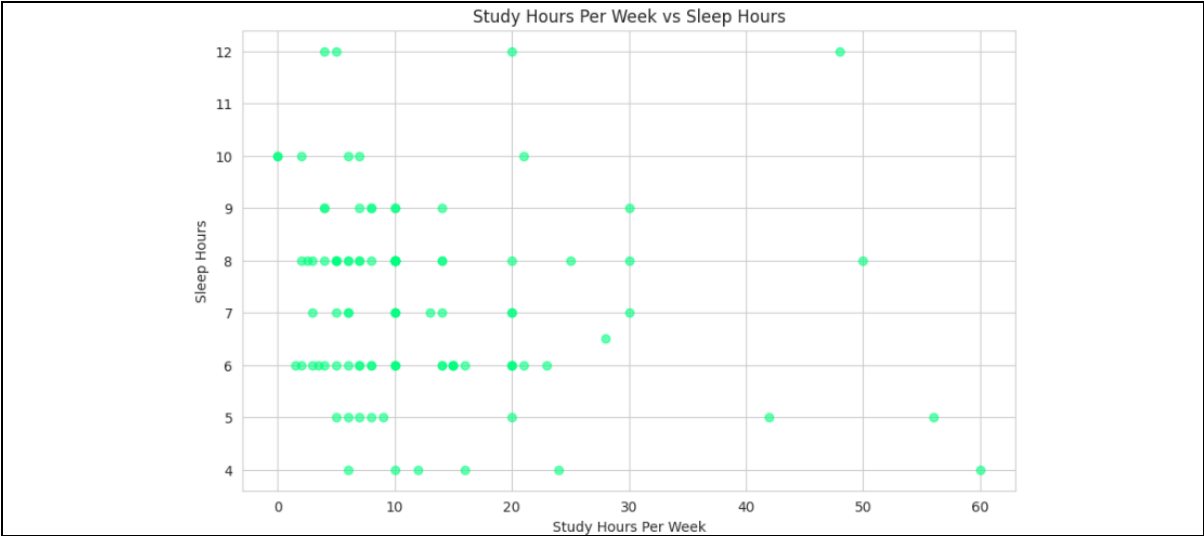
This scatter plot shows the relationship between the number of hours students study per week and their GPA. The data points appear spread out across the graph, indicating that there is **no strong linear relationship** between study hours and GPA in this dataset. While some students who study more hours tend to have slightly higher GPAs, many students with moderate study hours also achieve similar GPA values. This suggests that studying more hours alone may not directly determine academic performance.

Figure 8: Heatmap Showing the Density of Study Hours and GPA



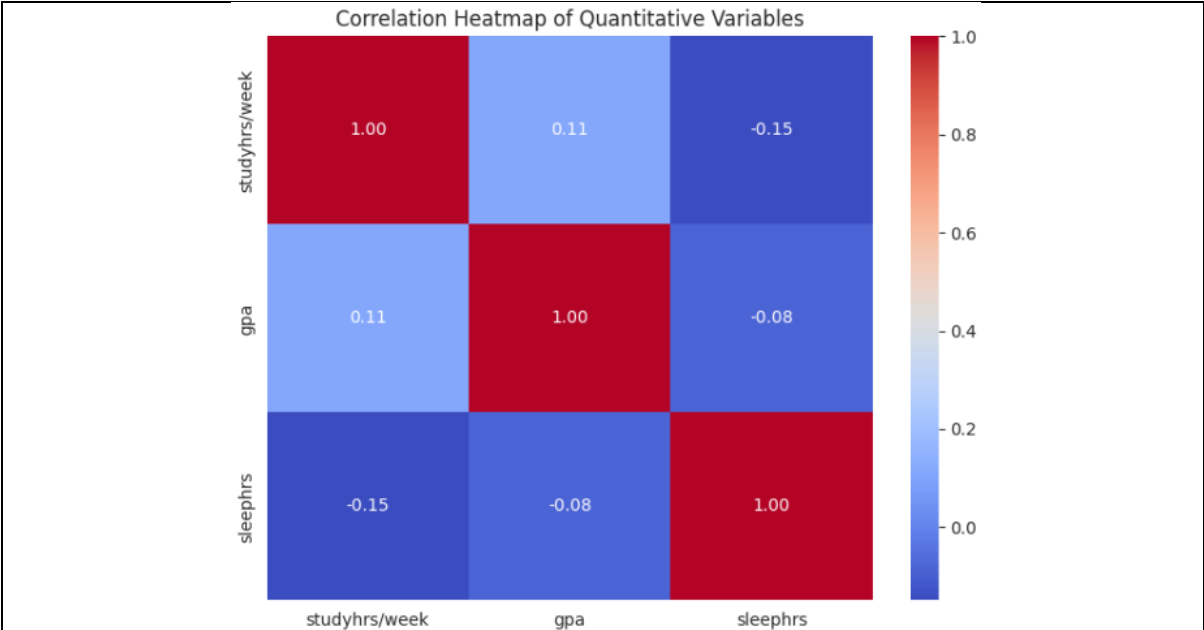
This heatmap visualizes the density of data points between weekly study hours and GPA. The brighter areas indicate where more students are concentrated within specific ranges of study hours and GPA values. Most students appear clustered around **5 to 15 study hours per week with GPAs between approximately 2.7 and 3.5**, showing that many students fall within this common range. The visualization also confirms that there is **no strong consistent pattern**, suggesting that study hours alone do not strongly determine GPA.

Figure 9: Scatter Plot Showing the Relationship Between Weekly Study Hours and Sleep Hours



This scatter plot illustrates the relationship between the number of hours students study per week and the number of hours they sleep per night. The data points are spread across the graph without forming a clear pattern, indicating that there is **no strong relationship between study hours and sleep duration**. Some students who study many hours still maintain normal sleep patterns, while others who study fewer hours report varying sleep durations. This suggests that study time does not necessarily reduce or increase the amount of sleep students get.

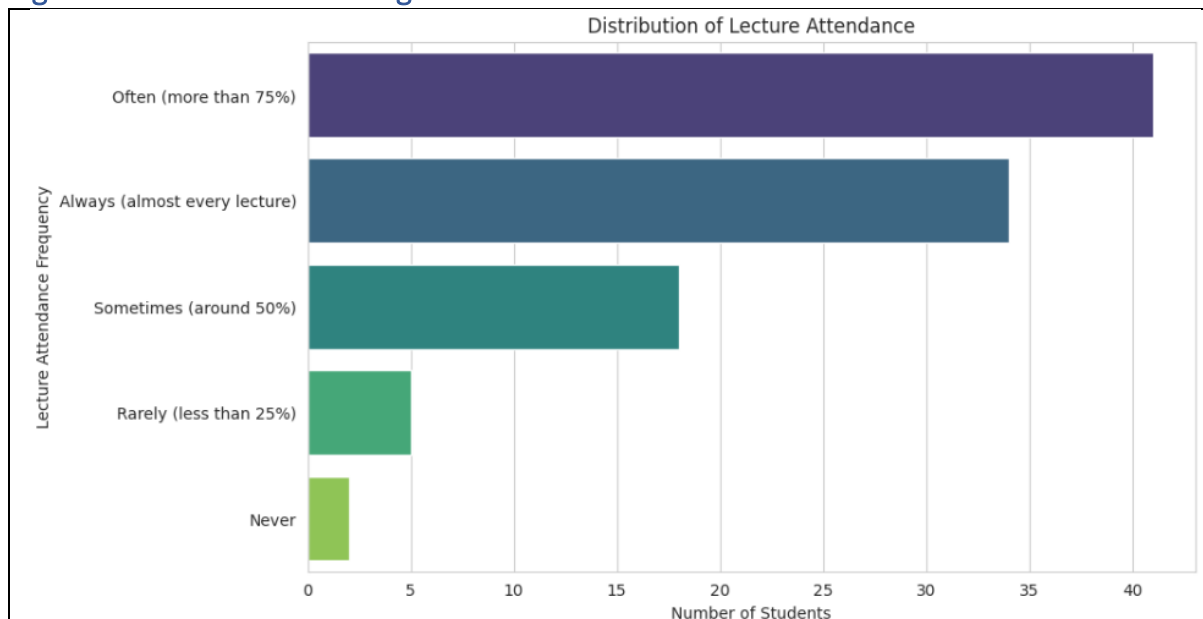
Figure 10: Correlation Heatmap of Quantitative Variables



This heatmap shows the correlation coefficients between the main quantitative variables: weekly study hours, GPA, and sleep hours. The values range between **-1 and 1**, where numbers closer to 1 or -1 indicate stronger relationships, while values close to 0 indicate weak or no linear relationship. The correlations in this dataset are all relatively close to zero, suggesting **weak relationships between the variables**. For example, the correlation between study hours and GPA is approximately **0.11**, indicating only a very weak positive relationship, while sleep hours show a slight negative correlation with both study hours and GPA.

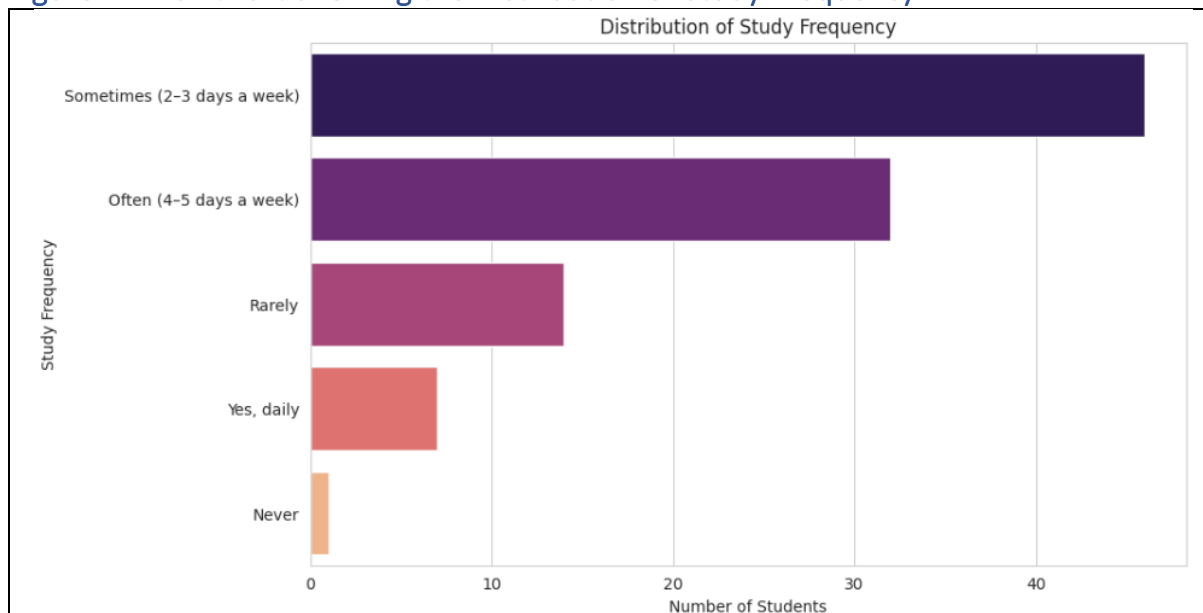
Part II: Qualitative Analysis

Figure 11: Bar Chart Showing the Distribution of Lecture Attendance



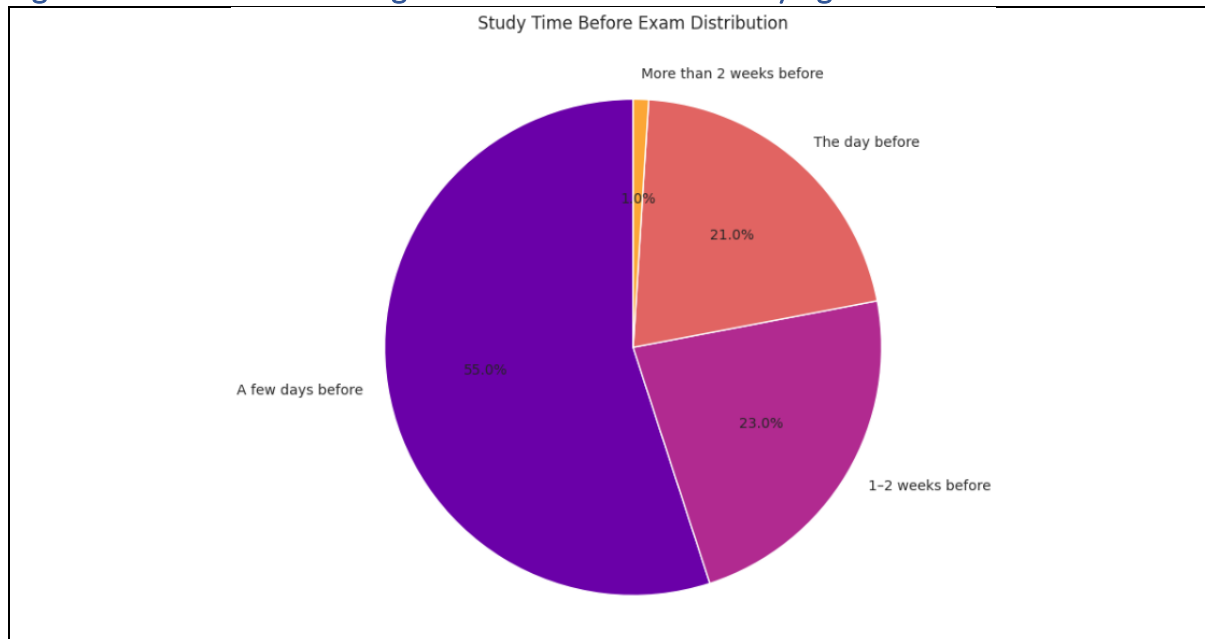
This bar chart shows how frequently students attend lectures based on the survey responses. The majority of students report attending lectures **often or always**, indicating that most participants maintain a high level of lecture attendance. A smaller number of students attend lectures only sometimes or rarely, while very few students report never attending lectures. This suggests that lecture attendance is generally high among the surveyed students.

Figure 12: Bar Chart Showing the Distribution of Study Frequency



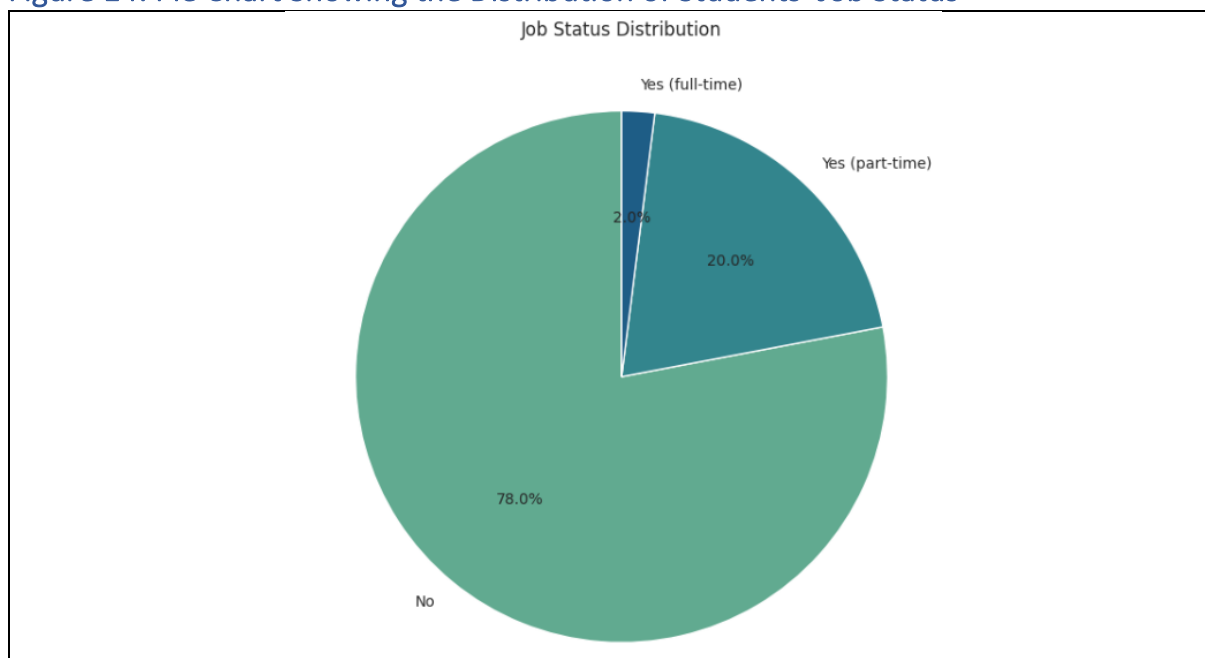
This bar chart illustrates how frequently students report studying during the week. The majority of students study **sometimes (2–3 days per week)** or **often (4–5 days per week)**, indicating that most participants maintain a moderate study routine rather than studying every day. A smaller number of students report studying daily, while very few students report never studying. This distribution suggests that regular but not necessarily daily studying is the most common pattern among the surveyed students.

Figure 13: Pie Chart Showing When Students Start Studying Before Exams



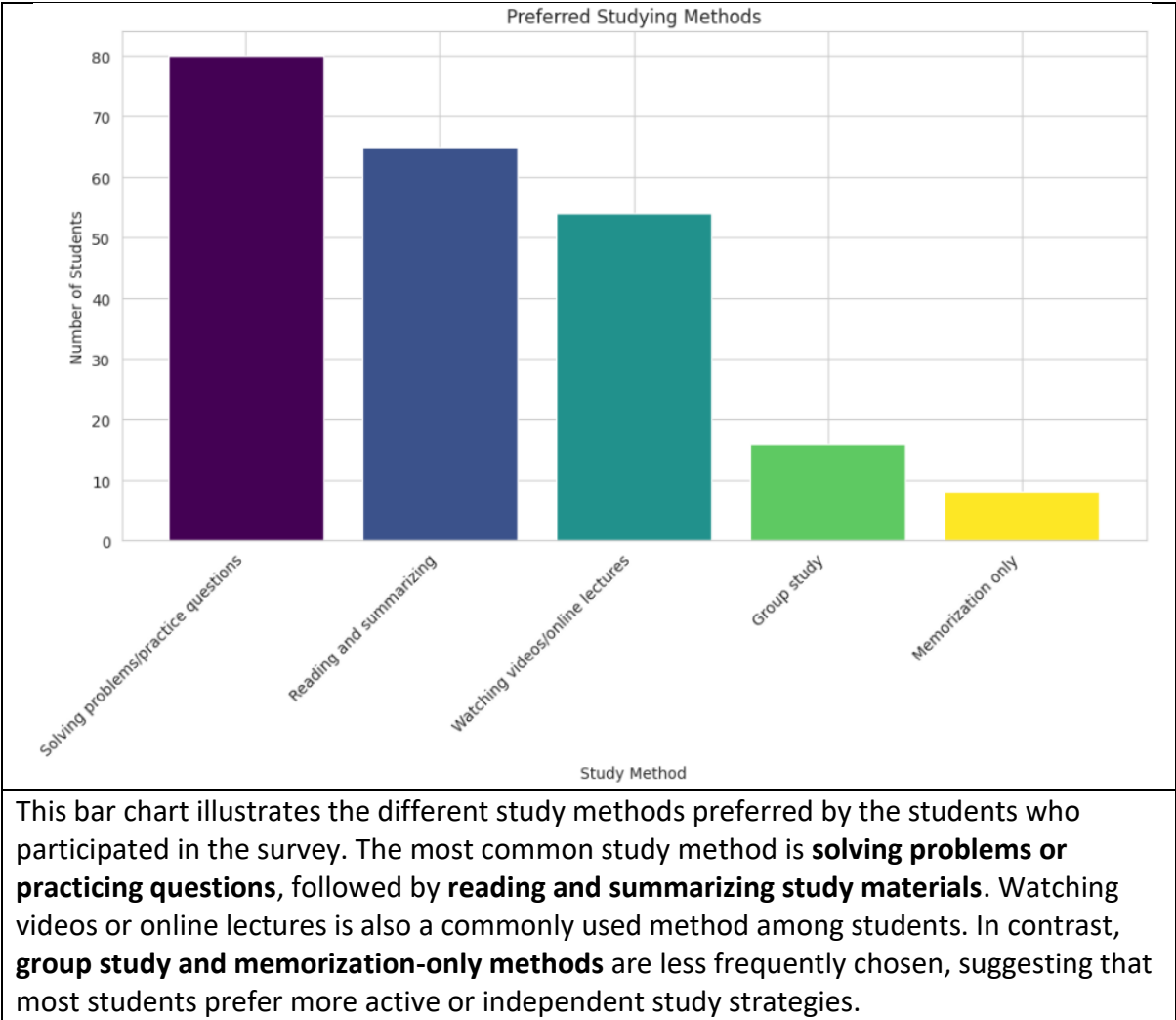
This pie chart illustrates when students typically begin studying before an exam. The majority of students, about **55%**, report that they start studying **a few days before the exam**, indicating that last-minute preparation is common among the participants. Smaller portions of students begin studying **one to two weeks before the exam (23%)** or **the day before (21%)**, while only a very small percentage start studying **more than two weeks in advance**. This suggests that most students tend to begin exam preparation relatively close to the exam date.

Figure 14: Pie Chart Showing the Distribution of Students' Job Status



This pie chart illustrates the employment status of the students who participated in the survey. The majority of students, about **78%**, reported that they do not currently have a job. Around **20% of students work part-time**, while only a very small percentage, about **2%**, work full-time. This distribution suggests that most students in the sample are primarily focused on their studies rather than employment.

Figure 15: Bar Chart Showing Students' Preferred Studying Methods



Part III: Analysis Summary

The visualizations presented in this analysis provide an overview of students' study habits, sleep patterns, attendance behavior, and academic performance. The distributions of study hours, sleep hours, and GPA show that most students fall within moderate ranges for these variables. Scatter plots and correlation analysis indicate that the relationships between study hours, sleep hours, and GPA are relatively weak in this dataset. Additionally, behavioral variables such as lecture attendance, study frequency, and exam preparation patterns provide insight into how students manage their academic responsibilities. Overall, the analysis suggests that academic performance is influenced by a combination of factors rather than a single variable.

Conclusion

This study aimed to explore the relationship between students' studying habits and their academic performance using data collected through a survey. Several variables were examined, including study hours, sleep hours, lecture attendance, study frequency, preferred study methods, and exam preparation habits. The analysis used descriptive statistics and visualizations to identify patterns and relationships within the dataset.

The results show that while students demonstrate a variety of study habits and behaviors, the correlation between study hours, sleep hours, and GPA appears to be relatively weak in this sample. This suggests that academic performance cannot be explained solely by the number of hours spent studying or sleeping. Instead, performance may be influenced by a combination of factors such as study strategies, time management, personal motivation, and external commitments.

Overall, the findings indicate that while certain study habits are common among students, there is no single behavior that strongly predicts academic performance in the dataset analyzed. Future studies with larger and more diverse samples may provide deeper insights into the factors that influence student success.

Critical Reflection

During the data collection process, several limitations were identified. First, the survey relied on self-reported responses, which may introduce inaccuracies if participants provided estimated or incorrect information. Second, the sample was collected using convenience sampling, which may not fully represent the broader population of university students. Additionally, the relatively small sample size limits the ability to generalize the findings to all students. Despite these limitations, the collected data still provides useful insights into students' studying habits and academic behaviors.

These findings highlight the importance of considering multiple behavioral and environmental factors when evaluating student academic performance.